🪴 Smart Plant Monitor

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CSC 494 • Final Presentation

The Problem

- People forget to water their plants
- Sometimes plants get too much or too little water
- There’s no simple way to monitor plant health in real time

The Solution

- Build an IoT plant monitoring system
- Track:
  - Soil moisture
  - Temperature
  - Humidity
- Send data wirelessly to a dashboard
- Show plant health in real time

My Tools & Technology

Hardware:

- XIAO ESP32C6
- Soil Moisture Sensor
- AHT10 Sensor
- Breadboard + wires

Software:

- Arduino (C++)
- Python (server)
- Streamlit (dashboard)

How It Works

1. Sensors collect data
2. ESP32 reads the data
3. Data is sent over WiFi
4. Python server receives it
5. Dashboard displays it live

Sprint Plan

Project is split into **two 5-week sprints**

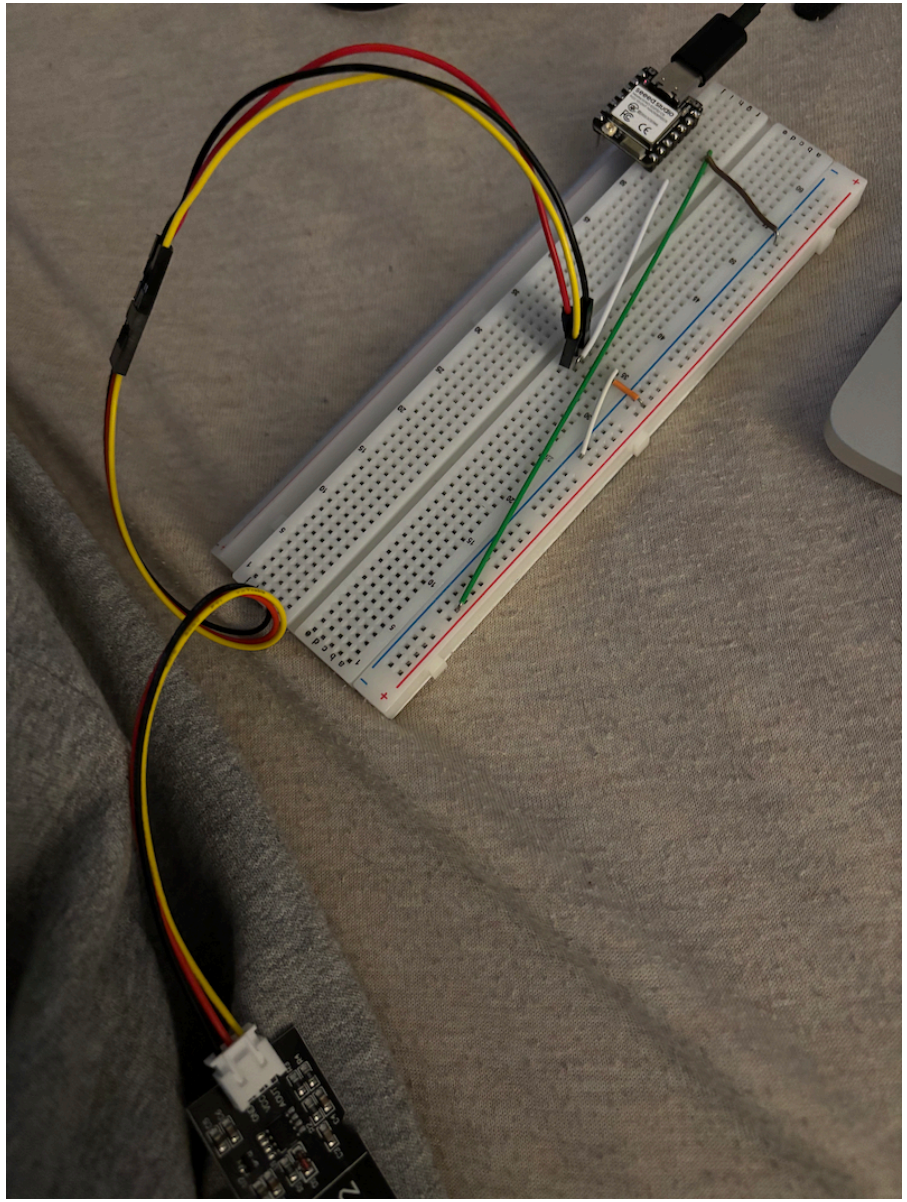
- Sprint 1 → Build foundation
- Sprint 2 → Add intelligence & polish

Sprint 1 – MVP & Connectivity

Goals:

- Set up hardware
- Read temperature & moisture
- Build basic dashboard (Arduino IDE)

First attempt at Breadboard

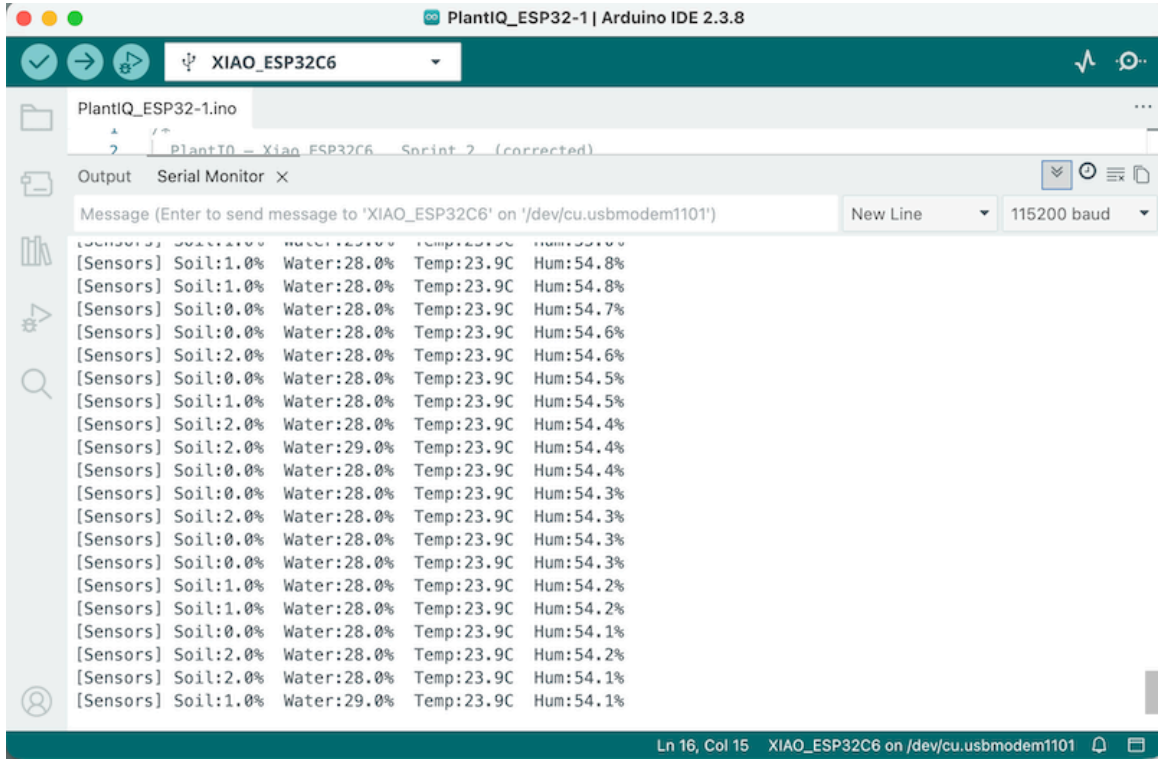


Sprint 1 Results

- Live sensor data working
- ESP32 successfully sending data

Example:

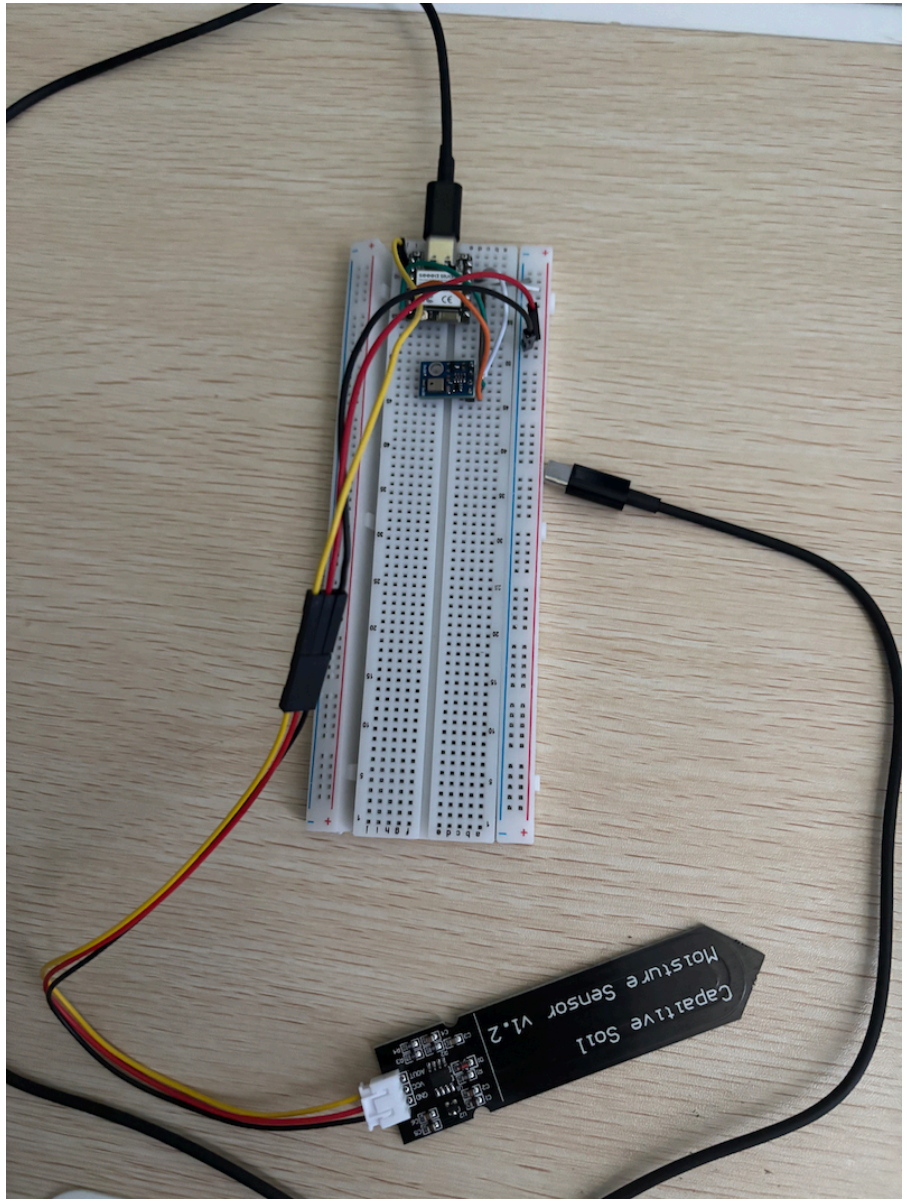
- Soil Moisture: 2%
- Temperature: 21.4°C
- Humidity: 43%



Sprint 1 board result

Data Visualization

- Added live graphs
- Tracks changes over time
- Helps understand plant conditions



Sprint 2 – Intelligence & Features

Goals:

- Connect ESP32 to WiFi
- Python server receiving data
- Send data to server
- Store historical data
- Improve dashboard UI
- Add watering recommendations
- Add alerts/notifications
- Dashboard showing real-time values
- Test and polish system

Learned with AI

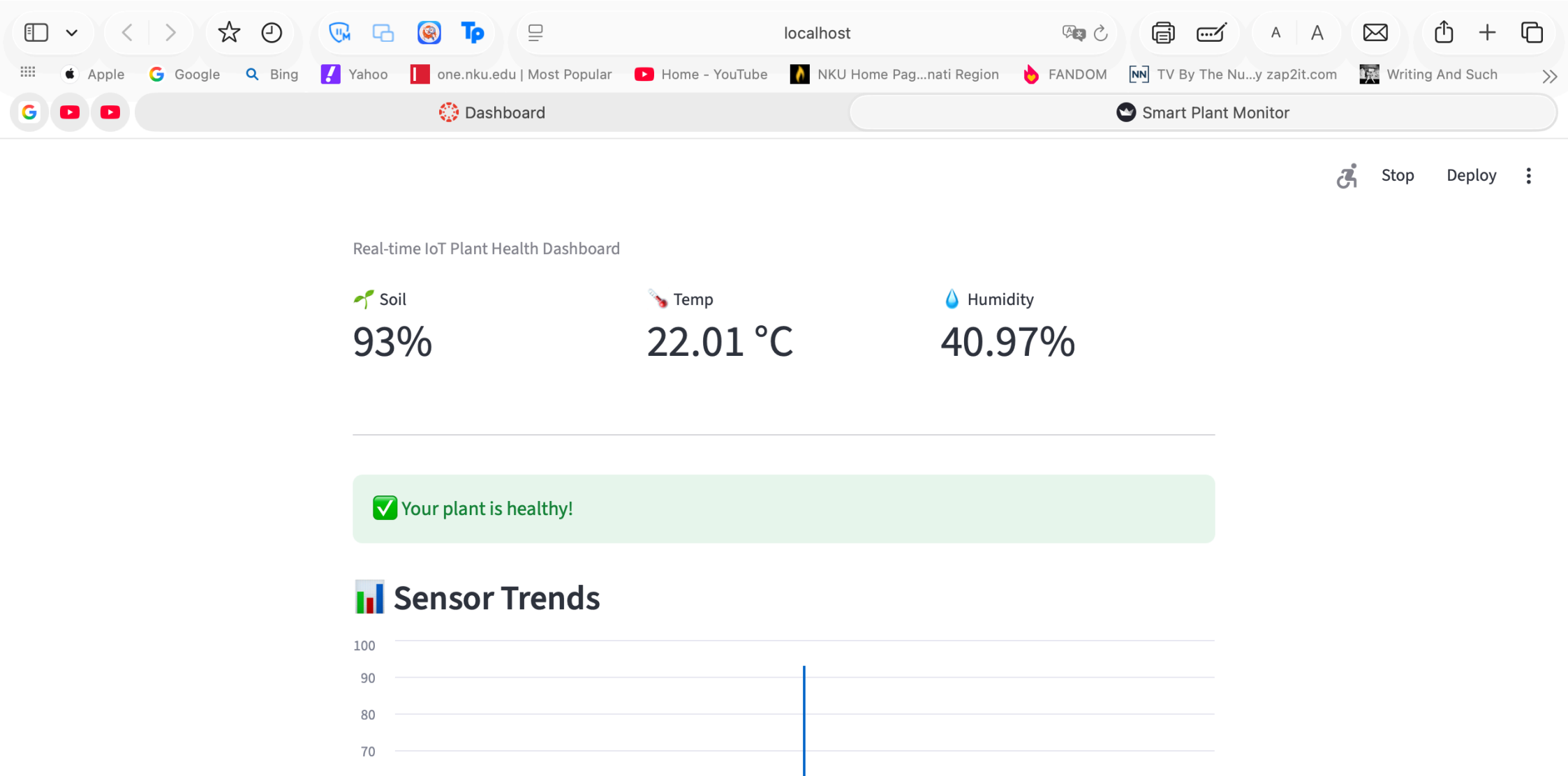
- Helped debug Arduino code
- Helped connect WiFi + server
- Helped build dashboard
- Helped understand full system

Learning with AI:

Topic 1: IoT Sensor Communication (HTTP vs MQTT)

Topic 2: SQLite for IoT Data

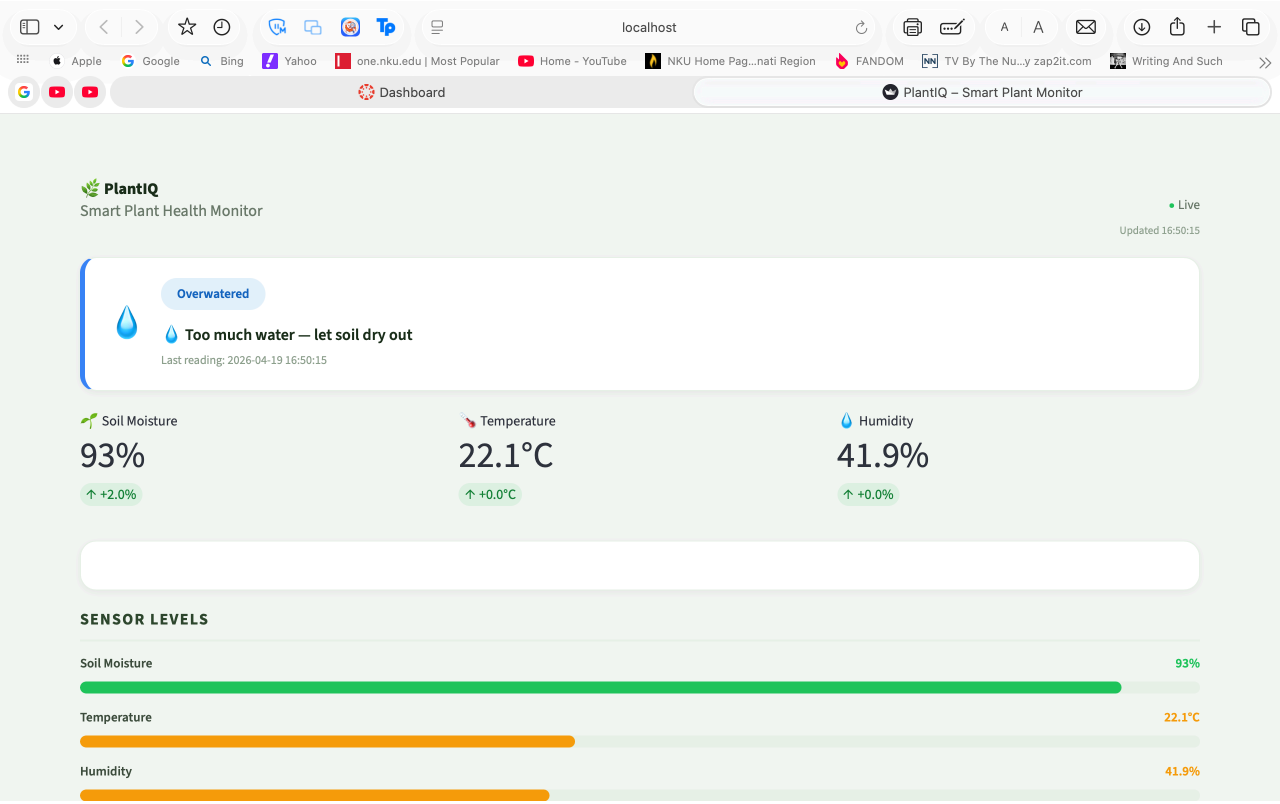
Dashboard visualization picture



Dashboard

- Built with Streamlit
- Displays:
  - Soil moisture
  - Temperature
  - Humidity
- Shows plant status:
  - Needs water
  - Healthy

Dashboard Picture



Demo (live)

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Video
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- YouTube

Some Challenges

Some Challenges

- No Automatic Pump
- Wiring sensors correctly
- Getting consistent sensor readings
- WiFi connection issues
- Connecting everything together

Future Improvements

- Add water pump automation
- Make dashboard look more like an app
- Add notifications
- Deploy online

Time of Reflection

- Learned how hardware and software connect
- Got better at debugging
- Built a full working IoT system

🙌🏾 Thank You

Questions?